MIL-M-63184(AR)
10 October 1980
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

MOUNT, RESILIENT, for M483 CONTAINER

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of vibration shock mount, designated Mount, Resilient, which is used in the Pershing M483 warhead section container.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8861928 - Carton, Packing, Ammunition, for Mount, Resilient.
10635480 - Mount, Resilient.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC NUOR

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials and components.- Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

3.2 Assembly.- The resilient mount shall comply with all requirements specified on drawing (dwg.) 10635480, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article.- When specified, the contractor shall submit sample resilient mounts for first article inspection and approval (see 4.3 and 6.1.1).

3.4 Transmissibility.- The transmissibility value of the mounts shall be not greater than 5 when subjected to the vibrations specified in Table I. Testing shall be as specified in 4.5.1.

TABLE I. Transmissibility vibration.

Frequency Range (cps)	Input	Number of Sweeps	Sweep Time (minutes)
5-10-5	0.13" da	2	1 max.
10-17-10	0.09" da	2	1 max.
17-26-17	1.3 g	2	1 max.
26-52-26	0.036" da	1	1 max.
52-300-52	5.0 g	1	2 max.

3.5 Load deflection.

3.5.1 Loading.- The resilient mounts shall be deflected as specified in Table II, and the corresponding load for each deflection shall be within the limits specified. During the deflection period, the mounts shall be visually inspected (to the extent possible) for cracks or tears in the elastomer and for separation of the elastomer from the metal plates. This visual inspection shall be repeated upon completion of this test (after removal of the mounts from the test fixture). In addition, during inspection testing only (4.4.3), the mounts shall exhibit no distortion of the elastomer which will result in (a) lack of parallelism of the mounting plates in excess of 5/8 inch and (b) displacement of the center of the round mounting plate in excess of 5/8 inch. These measurements shall be made within a 30-minute period after removal of the mounts from the test fixture. Testing shall be as specified in 4.5.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

TABLE II. Load deflection.

Deflection (inches)	Applied Load (pounds)
1.0	480 ± 95
2.0	1000 ± 200
3.0	1800 ± 360
4.0	2560 ± 510

3.5.2 Deflection inspection.- Each resilient mount, not subjected to the loading test (3.5.1), shall be tested to a minimum deflection of 4.0 inches to ensure quality of the elastomer and of the bond. The mount shall be held at the 4-inch deflection for a period not exceeding 10 seconds to allow time to visually examine the mount for defects. The mount shall withstand this test without any tearing or other damage of the elastomer, or without any breaking of the metal-to-elastomer bond. Testing shall be as specified in 4.5.2.2.

3.6 Drop test.- The resilient mounts, supporting a load of 250 ± 25 pounds, shall withstand a free fall of 18 inches minimum without any tearing or other damage of the elastomer, or without any breaking of the metal-to-elastomer bond. The impact shock shall be not greater than 10 g at room temperature, 8.5 g at 160°F, and 14 g at -33°F. Testing shall be as specified in 4.5.3.

3.7 High temperature.- The resilient mounts shall withstand a high temperature of 160° ± 5°F, for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage. While at the high temperature (at the end of the high-temperature exposure period), the mounts shall comply with the impact requirement of 3.6. After the mounts return to room temperature, they shall be inspected visually (including the visual examination in 3.6). Testing shall be as specified in 4.5.4.

3.8 Low temperature.- The resilient mounts shall withstand a low temperature of -33° ± 3°F, for a minimum period of 4 hours, without evidence of cracking, distortion, or other damage. While at the low temperature (at the end of the low-temperature exposure period), the mounts shall comply with the impact requirement of 3.6. After the mounts return to room temperature, they shall be inspected visually (including the visual examination of 3.6). Testing shall be as specified in 4.5.5.

3.9 Fatigue.- The resilient mounts shall be subjected to a constant vibration at the natural frequency of the pair of test mounts, at a double amplitude of 0.06 ± 0.006 inch, for a minimum period of 1 hour. There shall be no tearing or other damage of the elastomer or no breaking of the metal-to-elastomer bond. After this test, the mounts shall comply with the requirements of 3.4 and 3.5.1. Testing shall be as specified in 4.5.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

3.10 Workmanship.- All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission.- When the contract requires first article inspection, the contractor shall submit four consecutively produced resilient mounts to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection and testing to be performed.- The first article inspection samples shall be inspected, as applicable, for the defects specified in 4.4.2.1.1, using inspection methods contained therein, and for the applicable requirements of Section 3, which are listed in Table III, using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The resilient mounts shall be tested in the order specified in Table III. If any mount fails to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests is considered destructive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

TABLE III. First article tests.

Test	Requirement paragrapn
1. Transmissibility	3.4
2. Loading	3.5.1
3. Drop test	3.6
4. High temperature	3.7
5. Low temperature	3.8
6. Fatigue	3.9

4.4 Quality conformance inspection.

4.4.1 Inspection lot.- A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination.- A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table IV). Any defect found in the sample, selected for the major or minor classification, except for major defect 108, shall reject the lot. If any defect is found during inspection of major defect 108, the item shall be removed from the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

TABLE IV. Inspection lot sampling.

Major Defect Table		Minor Defect Table	
Lot Size	Sample Size	Lot Size	Sample Size
1-30	all	1-30	all
31-45	29	31-45	27
46-60	35	46-60	30
61-75	38	61-75	33
76-95	42	76-95	35
96-130	46	96-130	37
131-180	50	131-180	39
181-250	52	181-250	41
251-400	54	251-400	42
401 and over	56	401 and over	44

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.4.2.1 Classification of defects.- Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Mount, Resilient (see dwg. 10635480).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of mounting hole (4 holes)	Gage
102.	Diameter of mounting hole, min. (4 holes)	Gage
103.	Location of threaded hole	Gage
104.	Pitch diameter of threaded hole	Gage
105.	Minor diameter of threaded hole, max.	Gage
106.	Depth of thread in hole	Gage
107.	Evidence of loose or incorrect bonding of metal plate (2 plates)	Visual
108.	Cuts, cavities, cracks, or gouges in elastomer (including cored area) 1/	Visual
Minor:		
201.	Diameter of counterbore, min. (4 places)	Gage
202.	Protective finish missing or improper	Visual
203.	Marking missing, improper, or illegible	Visual
204.	Evidence of poor workmanship (see 3.9)	Visual

1/ Inspection of defect 108 is to be conducted on a 100 percent basis. Tears are permitted within the counterbore area (four places).

4.4.3 Inspection testing.- The contractor shall perform all of the tests specified in Table V. None of these tests is considered destructive, and samples so tested shall be returned to the lot.

TABLE V. Quality conformance tests.

Test	Test classification
1. Deflection inspection	Major
2. Loading	Major

4.4.3.1 Deflection inspection (see 3.5.2).- Each resilient mount, that is not selected for the loading test, shall be subjected to this test (see 4.5.2.2). If any mount fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.4.3.2 Loading (see 3.5.1).- Four resilient mounts shall be selected from every lot of 40 mounts produced, or portion thereof, and shall be subjected to this test (see 4.5.2.1). If any mount fails to comply with the applicable requirements, the lot shall be rejected.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests.- Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or specified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.3).

4.4.4.2 Minor defects and tests.- The acceptance inspection equipment, operating instructions, and calibration procedures used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

4.5 Methods of inspection.- The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, a Government-approved test fixture shall be used which shall consist of two resilient mounts, a suspended mass of 250 ± 25 pounds, and a rigid framework suitable for the tests being conducted. The suspended mass shall be of uniform cross section, or of a symmetrical design, in order to balance the load between the two mounts. The suspended mass shall be attached between the two mounts, and this assembly shall be bolted horizontally in the framework. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F).

4.5.1 Transmissibility.- A test fixture shall be prepared as specified in 4.5, and this assembly shall be mounted to a vibration machine. A vibration pickup shall be mounted on the top of the suspended mass, directly above the mass center of gravity, and on the vibration input platform. The pickups shall be accelerometers capable of accurate measurements in the frequency range of the test. The mounts shall be subjected to the vibrations specified in Table I along a vertical axis which is parallel to the metal plates of the mount. During the vibration period, measure and record the frequency and the amplification as follows: (a) at the natural frequency, (b) at approximately 3 cps increments from 5 to 26 cps, (c) at approximately 10 cps increments from 26 to 100 cps, and (d) at approximately 50 cps increments from 100 to 300 cps. Transmissibility is the ratio of the vibration reading (g output) on the suspended mass to the vibration reading (g input) on the vibration input platform. NOTE: The natural frequency of the system is the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

frequency at which the transmissibility ratio (output g/input g) is the greatest. The mounts shall comply with the requirements of 3.4 during this test.

4.5.2 Load deflection.

4.5.2.1 Loading.- A test fixture shall be prepared as specified in 4.5, except a Government-approved lightweight yoke or bar (material optional) shall be used in lieu of the 250 pound mass. The specified loads shall be centrally applied to the top of the yoke, and deflection measurements shall be made from any suitable reference point that lies in a plane that passes through the vertical center line of the test yoke. The mounts shall be preflexed twice to a deflection of 2.5 ± 0.12 inch by applying and removing the required load. This load, and subsequent loads, shall be applied at a rate of 0.5 to 1.0 inch per minute. A preload (including yoke weight) of 50 pounds shall be applied to the pair of mounts prior to the third flexing (the start of testing). The load and deflection indicators shall be zeroed at this preload value (all readings shall be referenced from this point). The mounts shall be loaded, at the previously specified rate, to obtain the deflections specified in Table II. There shall be no interruption in the application of the load, and the exact value of the load, obtained for each specified deflection, shall be recorded. Upon completion of this test (after the last deflection point has been reached), the load shall be gradually removed from the yoke. The mounts shall comply with the requirements of 3.5.1 during this test.

4.5.2.2 Deflection inspection.- A test fixture shall be prepared as specified in 4.5, except a Government-approved lightweight yoke or bar (material optional) shall be used in lieu of the 250 pound mass. Apply a load (at any suitable rate) to deflect the mounts 4 inches from their unloaded reference point. The method of load application and the method of deflection measurements shall be the same as used during the loading test (4.5.2.1). The mounts shall not be preflexed or preloaded before this test. The mounts shall comply with the requirements of 3.5.2 during this test.

4.5.3 Drop test.- A test fixture shall be prepared as specified in 4.5, except suitable hardwood skids (oak or equivalent) shall be mounted to the base of the fixture to act as the impact surface. Vertical guide rails (or other suitable means) shall be used to prevent tilting of the fixture during the drop test. An accelerometer shall be mounted on top of the suspended mass, directly above the mass center of gravity, to measure the deceleration forces on impact. The test fixture shall be raised to a height of 18 inches and released for a free fall onto a concrete or similar hard surface of sufficient mass to absorb the shock without deflection. The mounts shall comply with the requirements of 3.6 during this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.5.4 High temperature.- The test fixture used for the drop test (see 4.5.3) shall be used for this test. The fixture shall be placed into a test chamber, and the temperature of the chamber shall be raised to 160°F. The fixture shall remain at this temperature for a minimum period of 4 hours, and it shall then be tested as specified in 4.5.3 at the end of the high-temperature exposure period. After the mounts have been removed from the test fixture, they shall be visually inspected. The mounts shall comply with the requirements of 3.7 during this test.

4.5.5 Low temperature.- The test fixture used for the drop test (see 4.5.3) shall be used for this test. The fixture shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The fixture shall remain at this temperature for a minimum period of 4 hours, and it shall then be tested as specified in 4.5.3 at the end of the low-temperature exposure period. After the mounts have been removed from the test fixture, they shall be visually inspected. The mounts shall comply with the requirements of 3.8 during this test.

4.5.6 Fatigue.- A test fixture shall be prepared as specified in 4.5, and this assembly shall be mounted to a vibration machine. A vibration pickup shall be mounted on top of the suspended mass, directly above the mass center of gravity. The mounts shall be subjected to a constant vibration, at the natural frequency of the system (as determined during the transmissibility test (see 4.5.1)), at a double amplitude of 0.06 inch. The mounts shall be vibrated for a minimum period of 1 hour along a vertical axis which is parallel to the metal plates of the mount. The mounts shall comply with the requirements of 3.9 during this test.

5. PACKAGING

5.1 Preservation, packaging, and marking.- The resilient mounts shall be preserved, packaged, and marked in accordance with dwg. 8861928.

5.2 Packing and marking.- The resilient mounts shall be packed and marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

6.1.1 Procurement requirements.- Title, number, and date of this specification and first article sample requirements.

6.1.2 Contract data requirements.- Contractor engineering documentation specified in 4.4.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

6.2 Objective evidence.- Records of the contractor's quality control and inspections which can be verified.

6.3 Acceptance inspection equipment.- The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Drawings and publications.- Drawings and publications listed in Section 2 of this specification under the heading of US Army Armament Research and Development Command (ARRADCOM) may also include drawings and publications prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings and publications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Supersession data.- This specification includes the requirements of Purchase Description PA-PD-2465, dated 7 April 1964, with Amendment 1 dated 14 December 1965.

Custodian:
Army-AR

(Project NUOR-A087)

Preparing activity:
Army-AR

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-63184(AR) Mount, Resilient for M483 Container

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAN-Q
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.